

Sri Lanka Institute of Information Technology

B.Sc. (Hons) Information Technology



Specialization in Information Technology

IT3010 –Network Design & Management

Laboratory Report 01

Student ID: IT23421226

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Introduction

Virtualization is a technology that enables multiple operating systems to be run on a single physical computer through the use of virtual machines, with each virtual machine operating as a separate computer equipped with its own operating system, storage, memory, and network settings. In this lab exercise, virtualization software such as VMware Workstation or Oracle VirtualBox was utilized to set up two virtual machines, where a Fedora desktop operating system was configured on the first machine and a CentOS server operating system was installed on the second, representing different use cases for desktop and server environments. Additionally, a virtual switch was created within the virtualization platform to enable communication between the two virtual machines, allowing them to interact as if they were physically connected and demonstrating how virtualization can be used to build isolated yet connected network environments for testing and learning purposes.

Objective

- Understand the concept of visualization
- Create virtual machines using a virtualization platform
- Install two operating system
 - Fedora
 - CentOS
- Configure a virtual network switch to connect both Virtual Machines

Theory–Virtualization

Virtualization refers to separating computing services from the physical hardware. Instead of using separate physical machines, multiple virtual machine systems can run simultaneously on a single host machine.

Benefits:

- Efficient hardware utilization
- Isolation between operating system
- Cost reduction
- Easy testing and deployment

Desktop OS vs Server OS

Desktop Operating System			Server Operating System		
	Features	Examples		Features	Examples
Installed on personal computers for everyday use.	- User friendly graphical interface - Application support (multimedia, office, browsing) - Basic security and automation	- Windows 7 - macOS - ubuntu - Desktop - Fedora	Installed on servers to manage networks, services and data.	- Strong security and stability - Advanced networking capabilities - Optional or minimal GUI	- Windows Server - Ubuntu Server - CentOS

Practical Procedure

Part A – Creating a Virtual Switch

1. Open VMware Workstation.
2. Go to **Edit → Virtual Network Editor**.

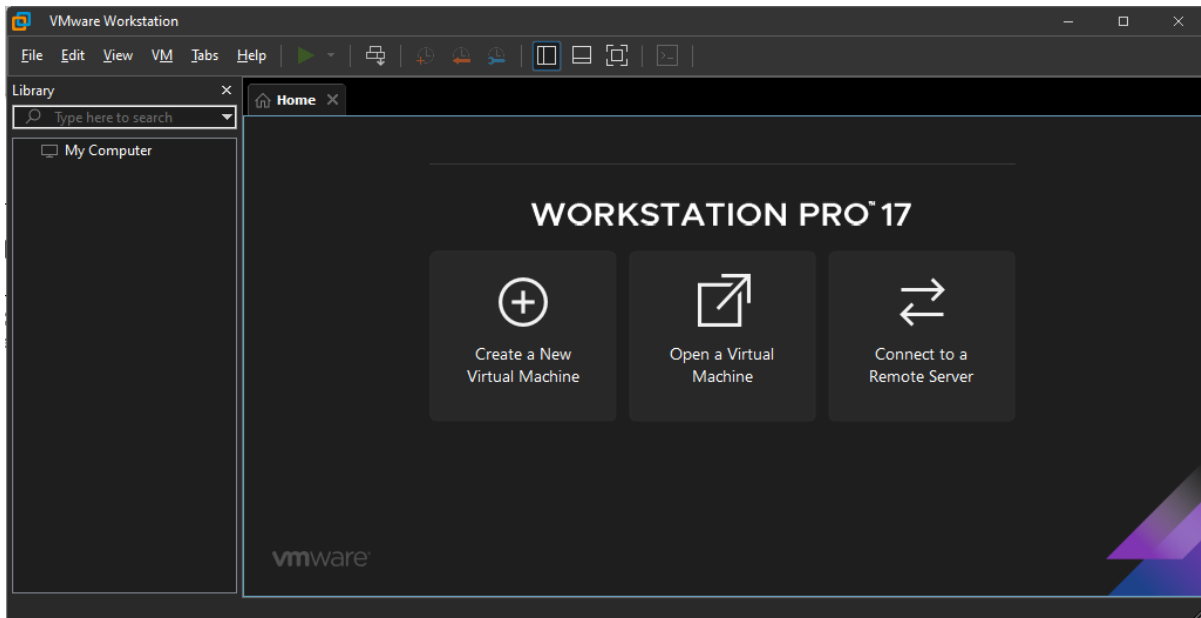


Figure 1 workstation pro-17

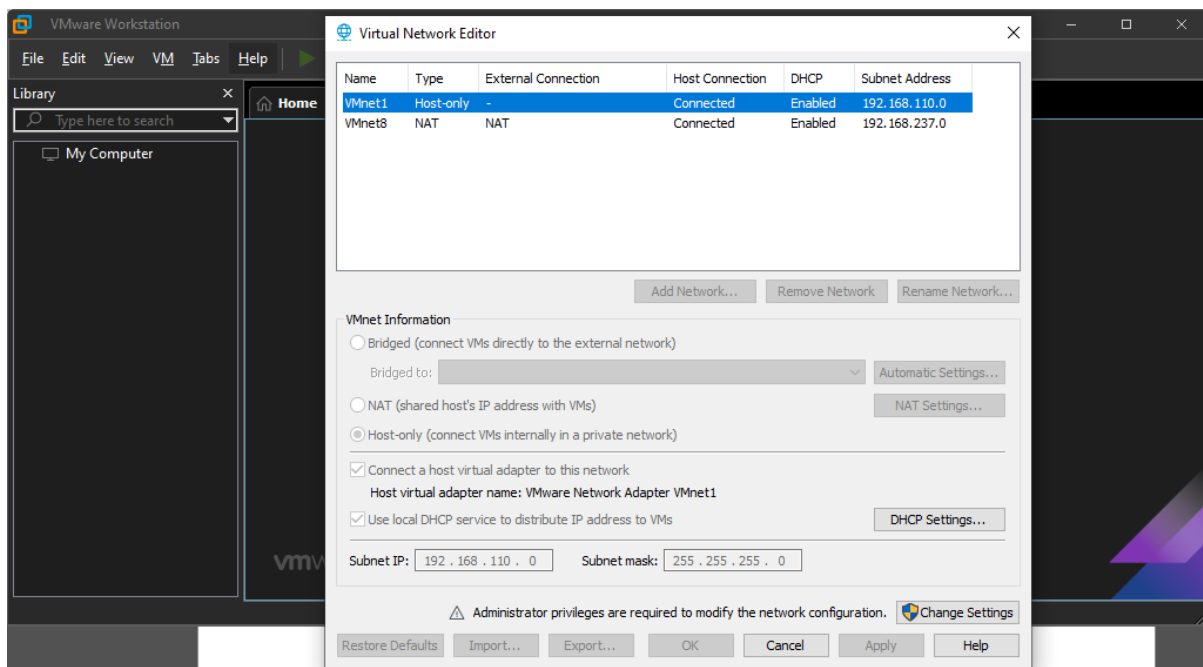


Figure 2 network editor

- Click **Change Settings** (administrator permission may be required).

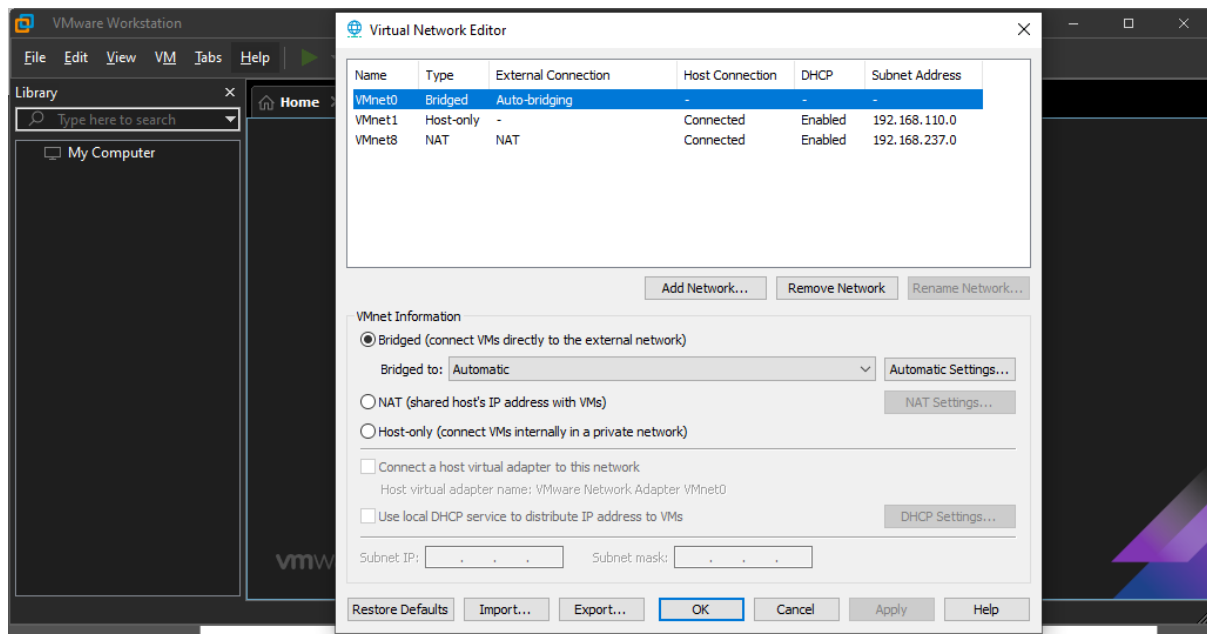


Figure 3 change setting

- Click **Add Network**.

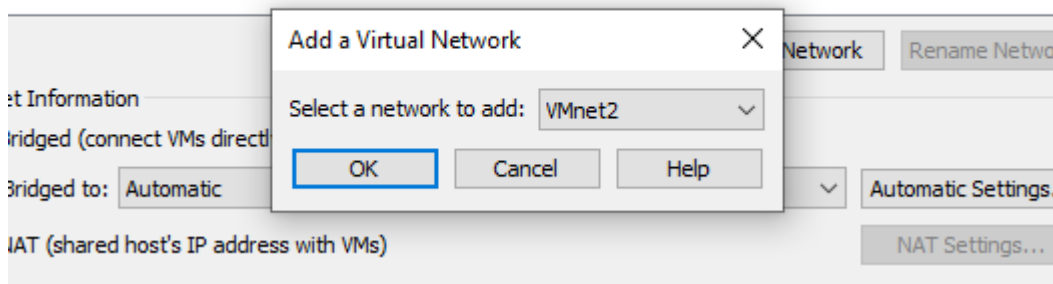


Figure 4 add network

5. Select **VMnet2** and confirm.

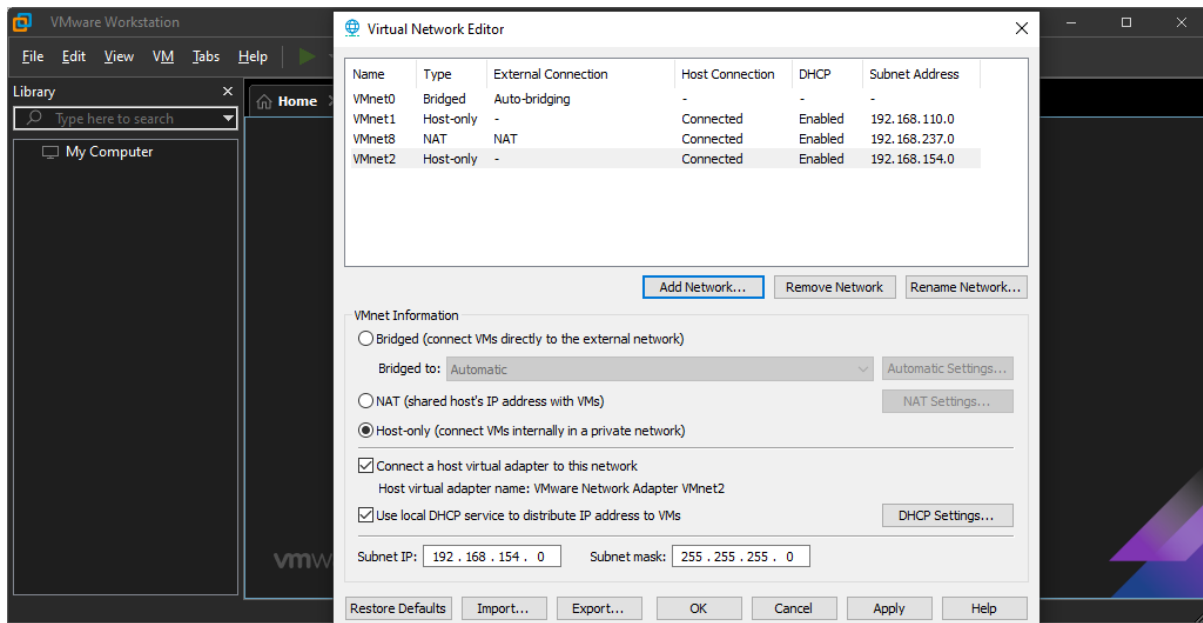


Figure 5 add VMnet2

6. Adjust network parameters if necessary.

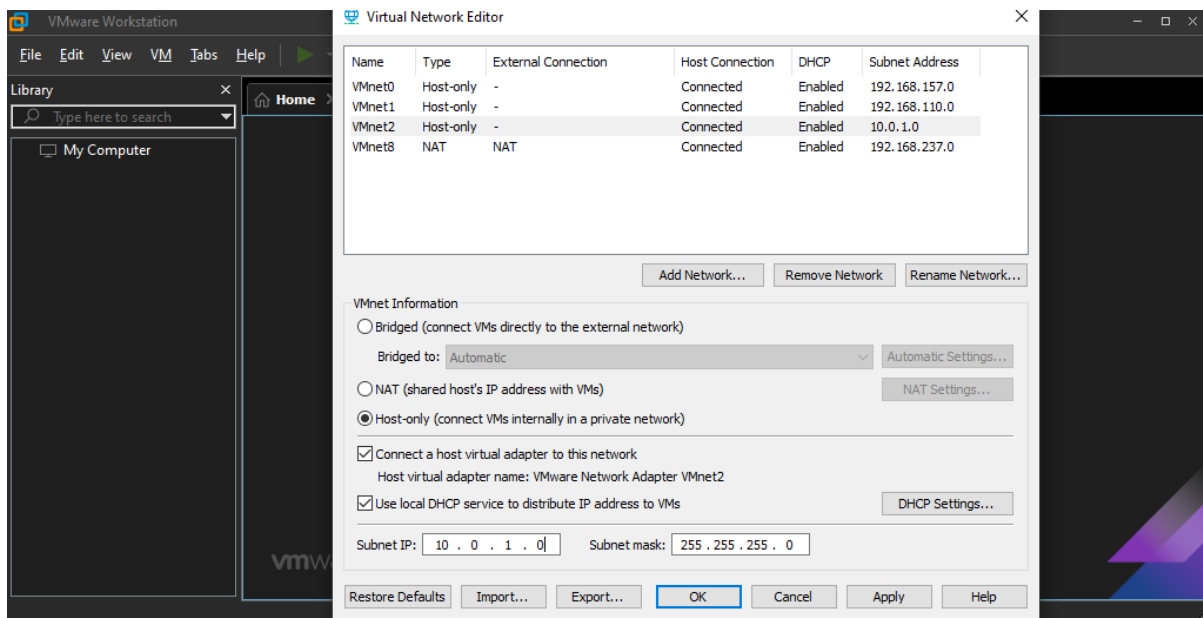


Figure 6 add network parameter

Part B – Creating Virtual Machines

Two VMs were created:

- Client VM → Fedora Desktop
- Server VM → CentOS Linux

1. Click **Create New Virtual Machine**.

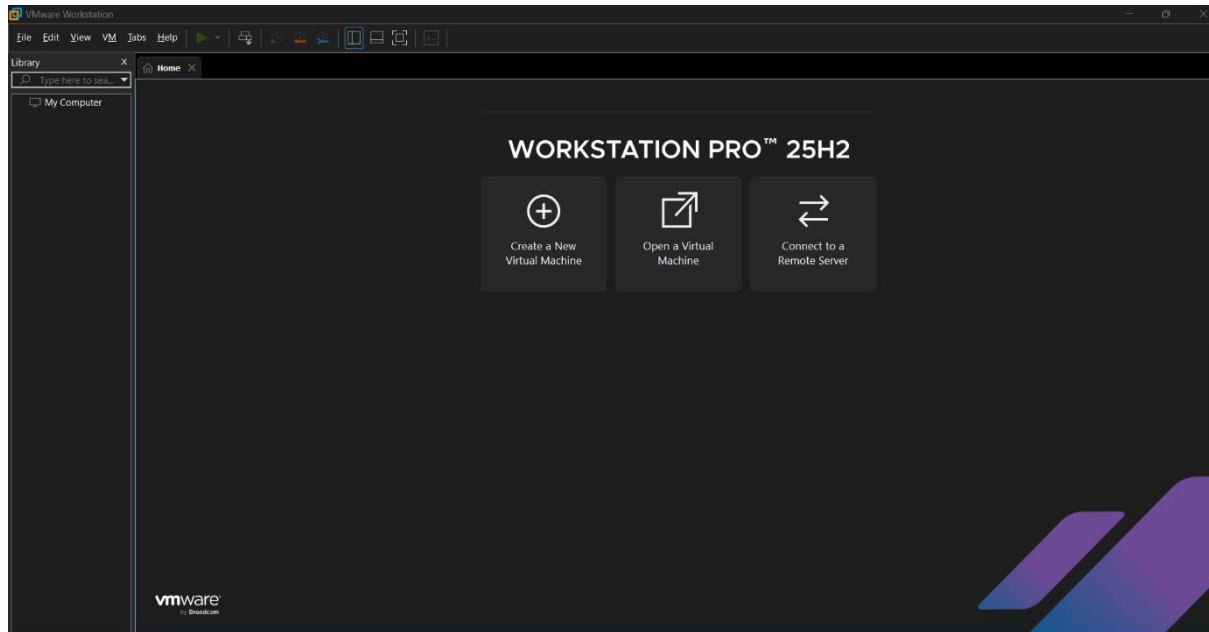


Figure 7 create new virtual machine

2. Select **Custom (Advanced)** setup.

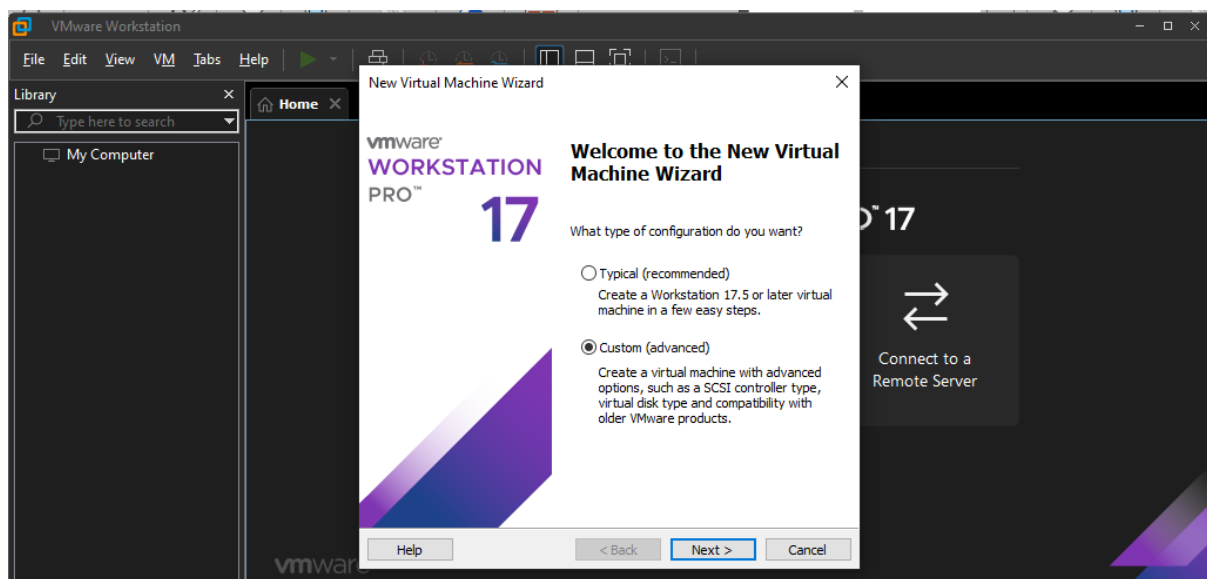


Figure 8 custom setup

3. Keep default compatibility settings.
4. Choose ISO installation image.

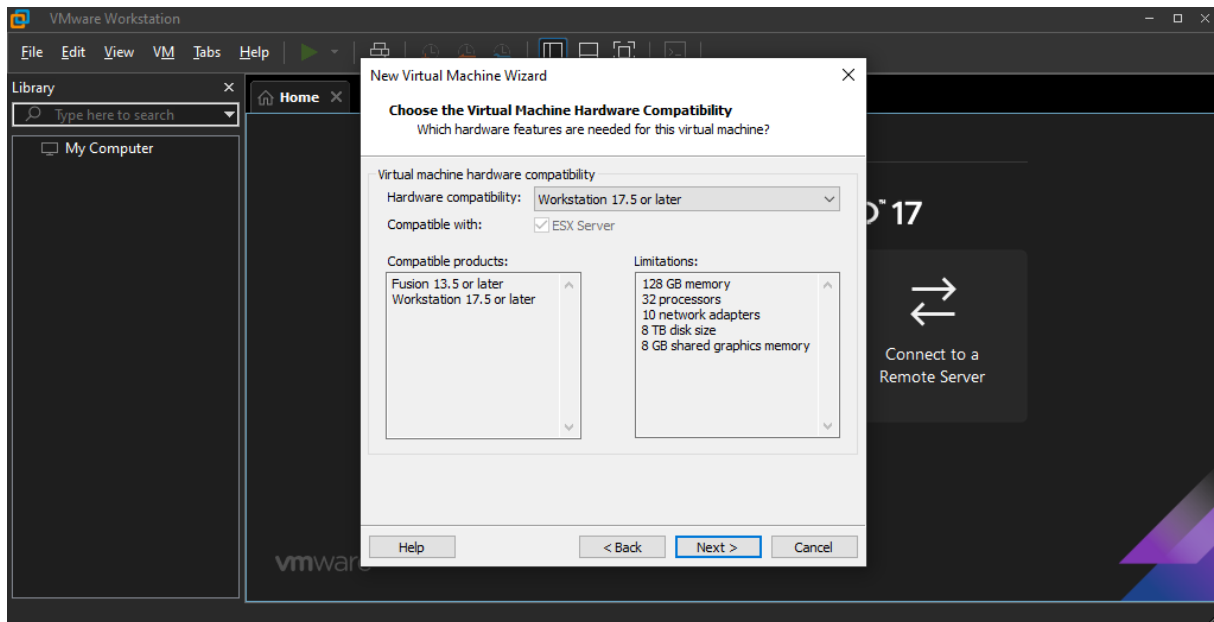


Figure 9 new virtual machine wizard

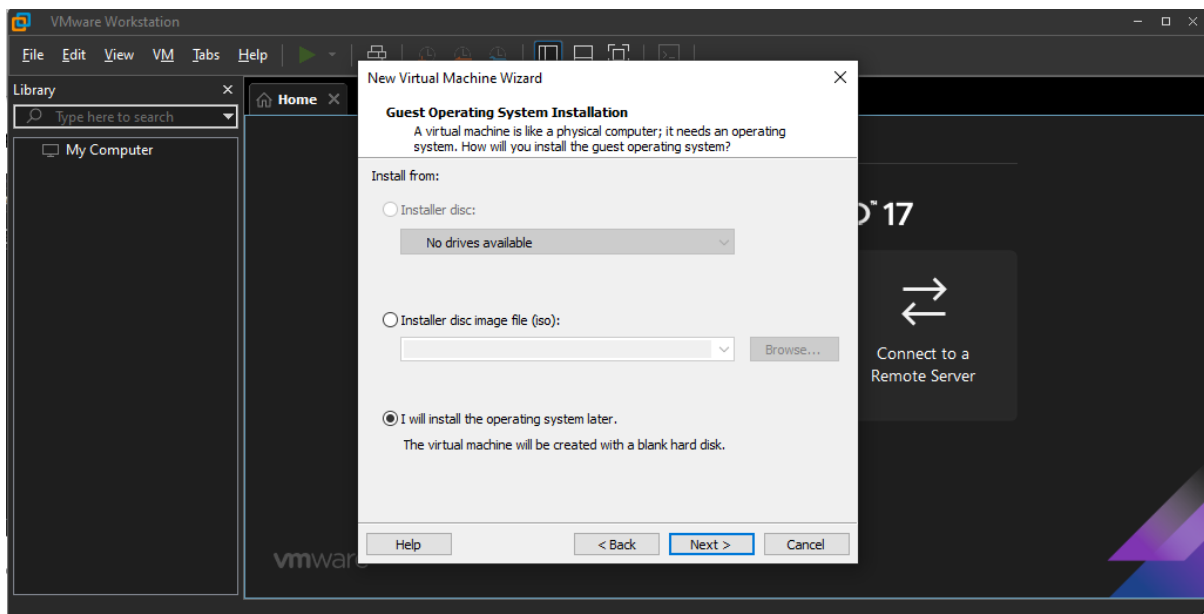


Figure 10 choose ISO install image

5. Select Linux OS type accordingly.

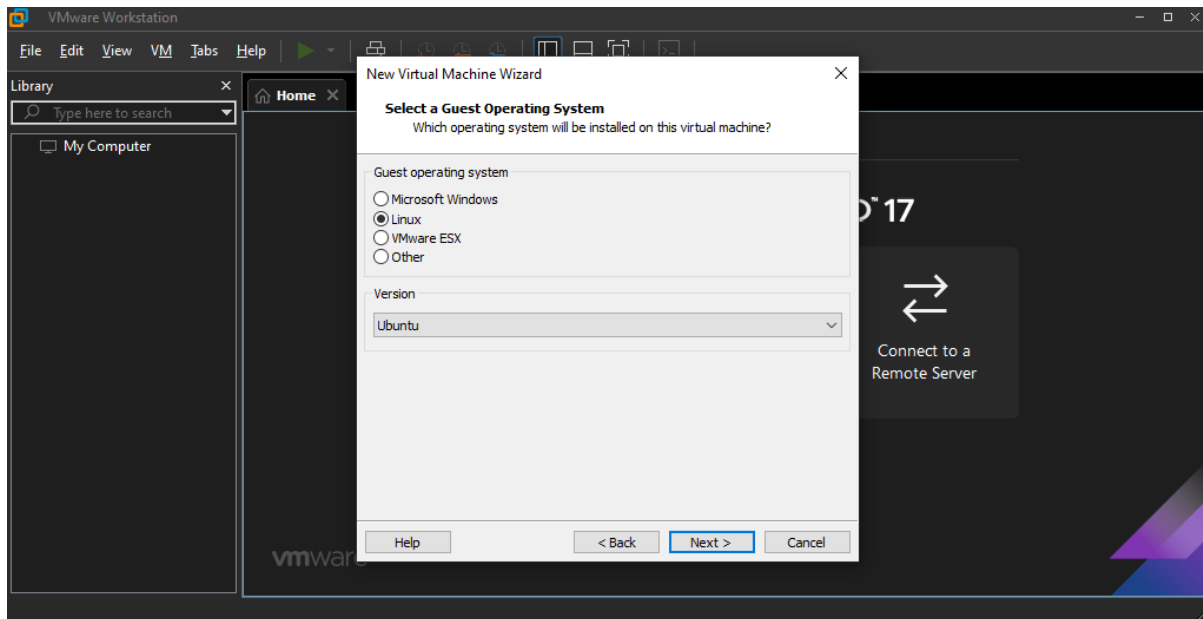


Figure 11 select Linux OS

6. Create storage folders

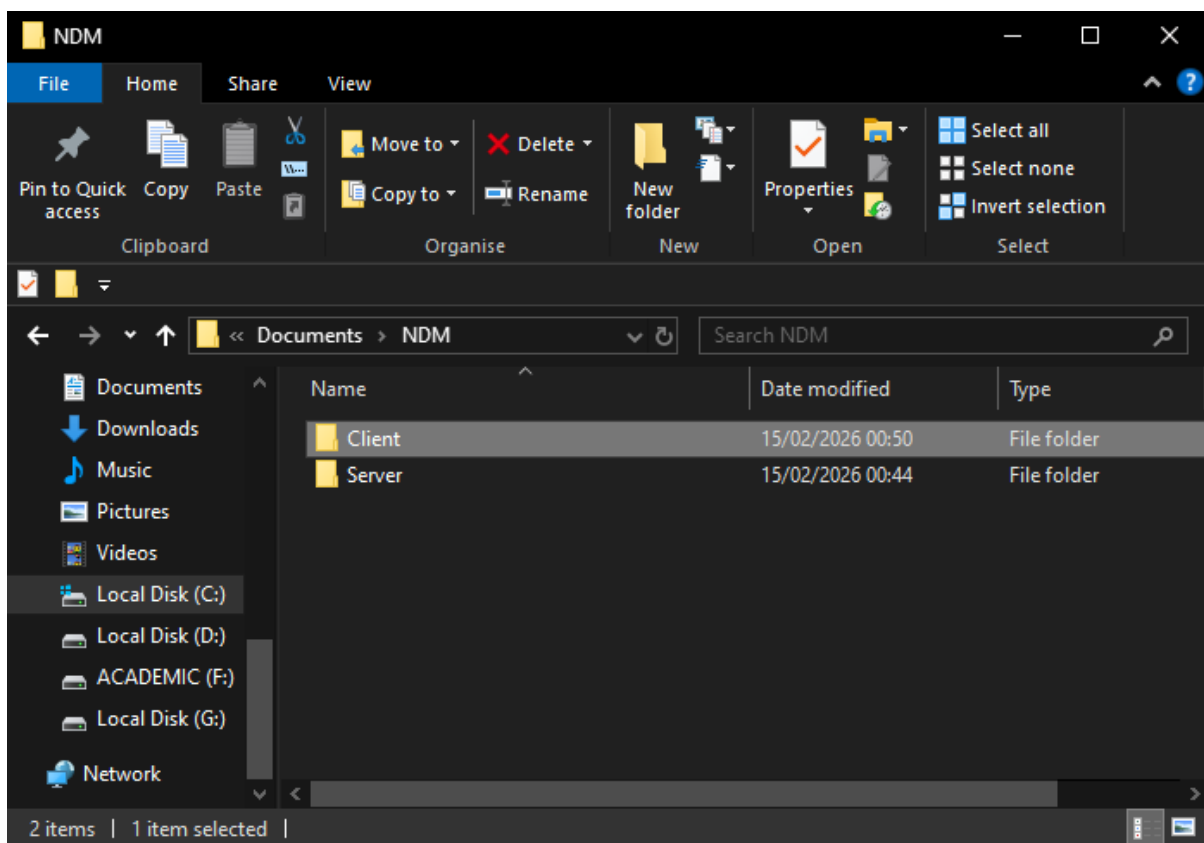


Figure 12 create storage folder

7. Allocate RAM, CPU, and disk space (default values acceptable).

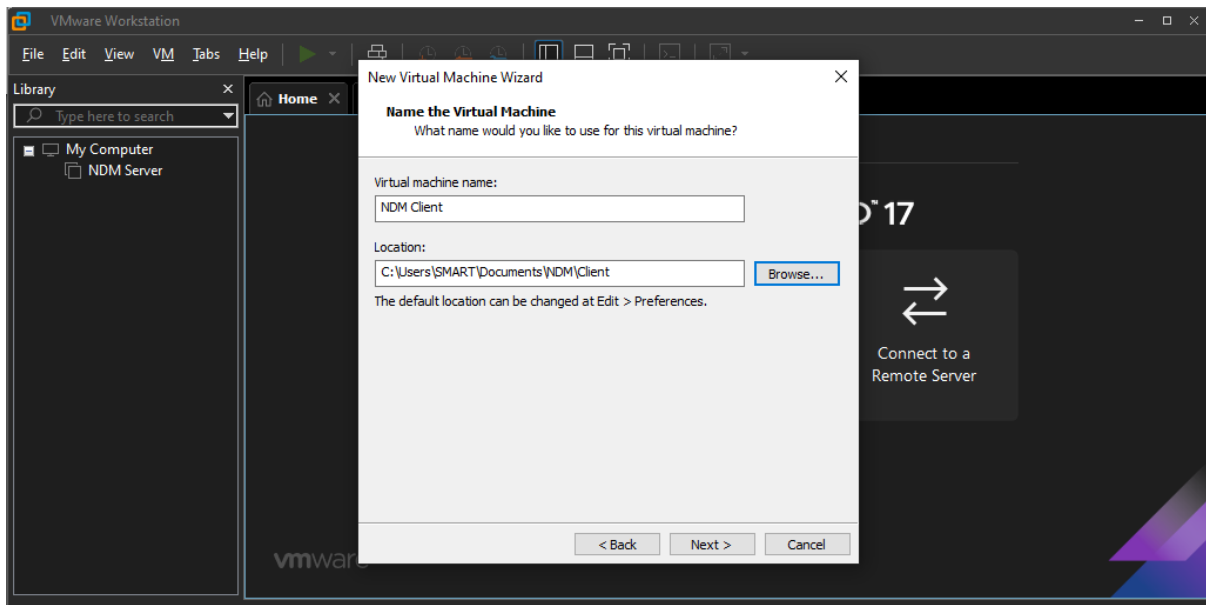
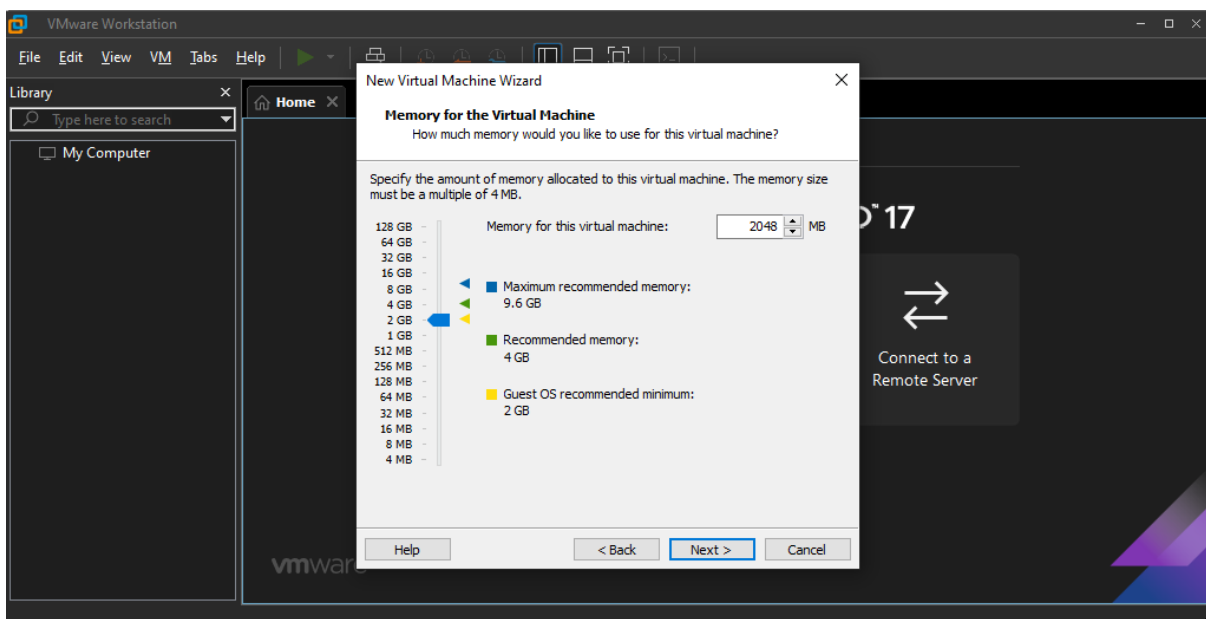
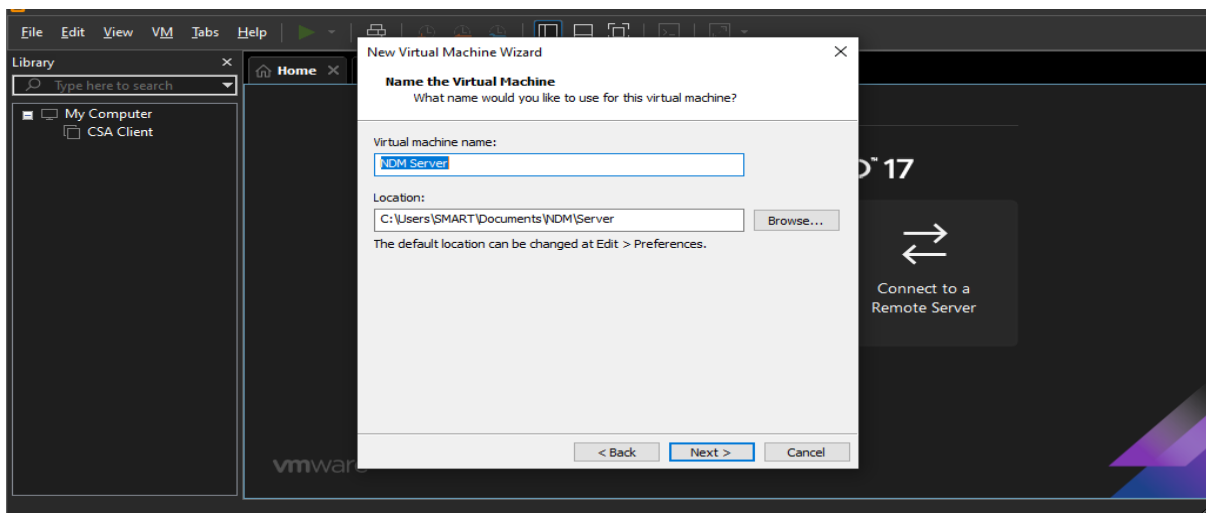
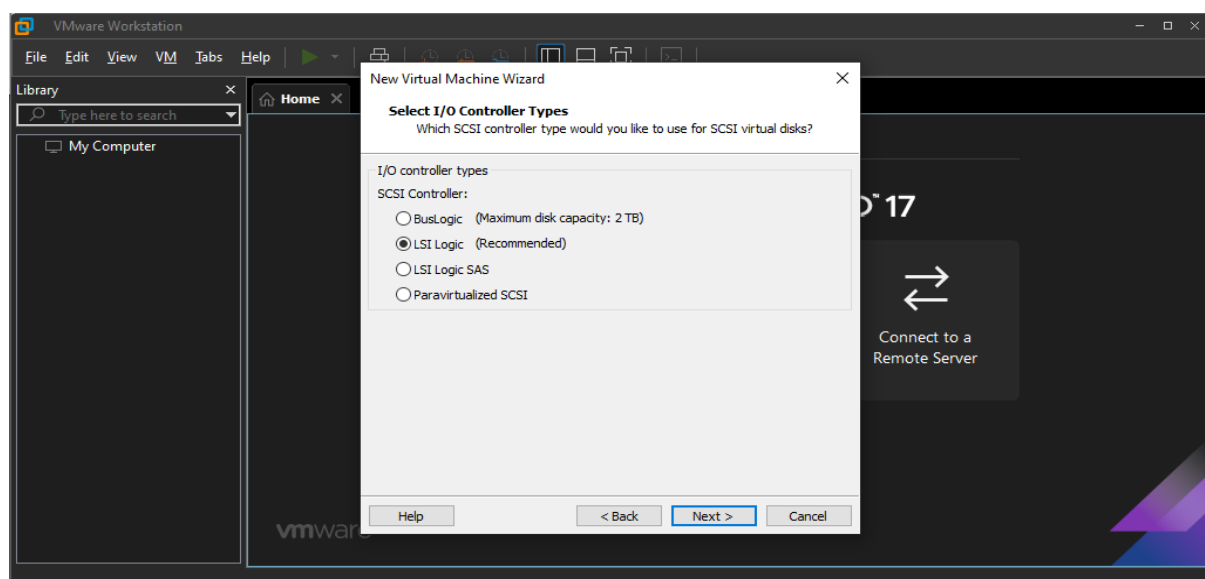
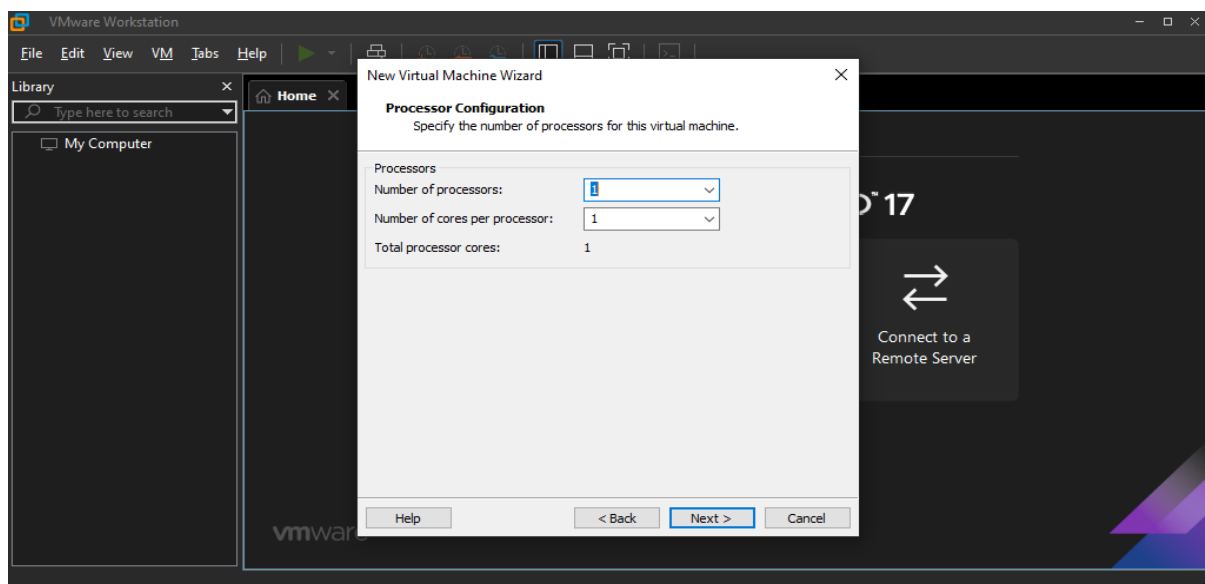
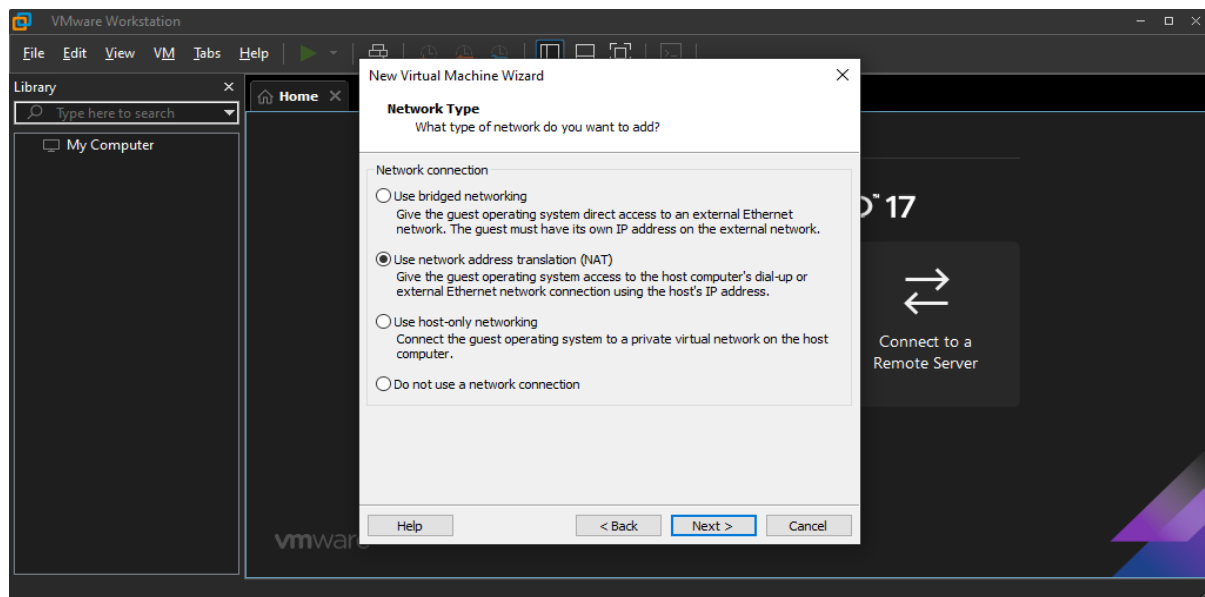
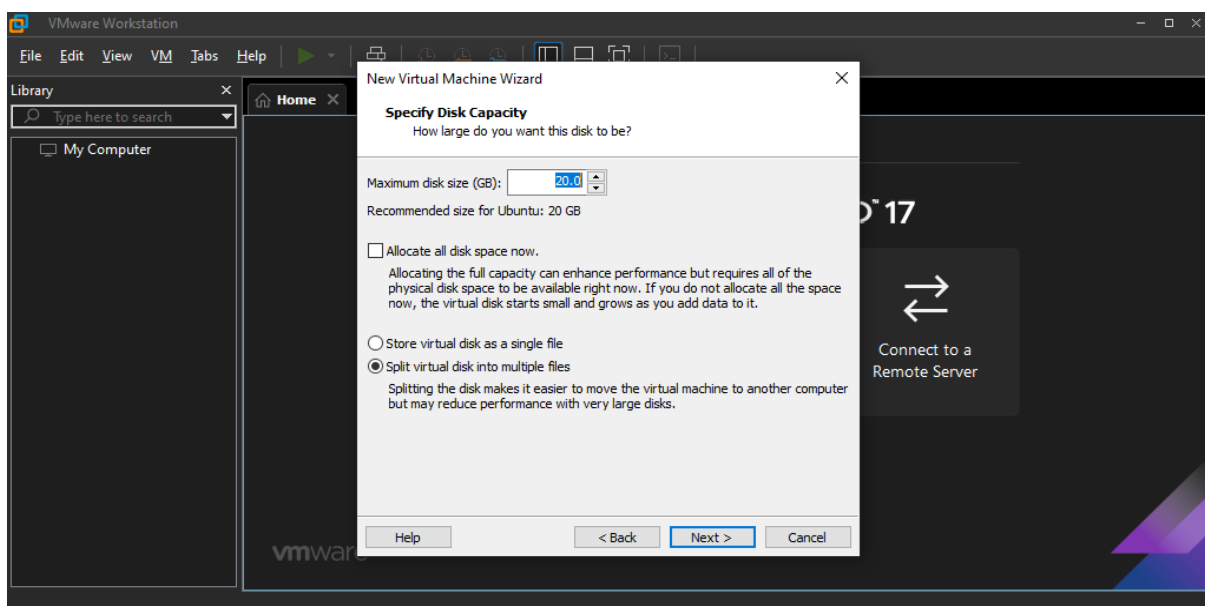
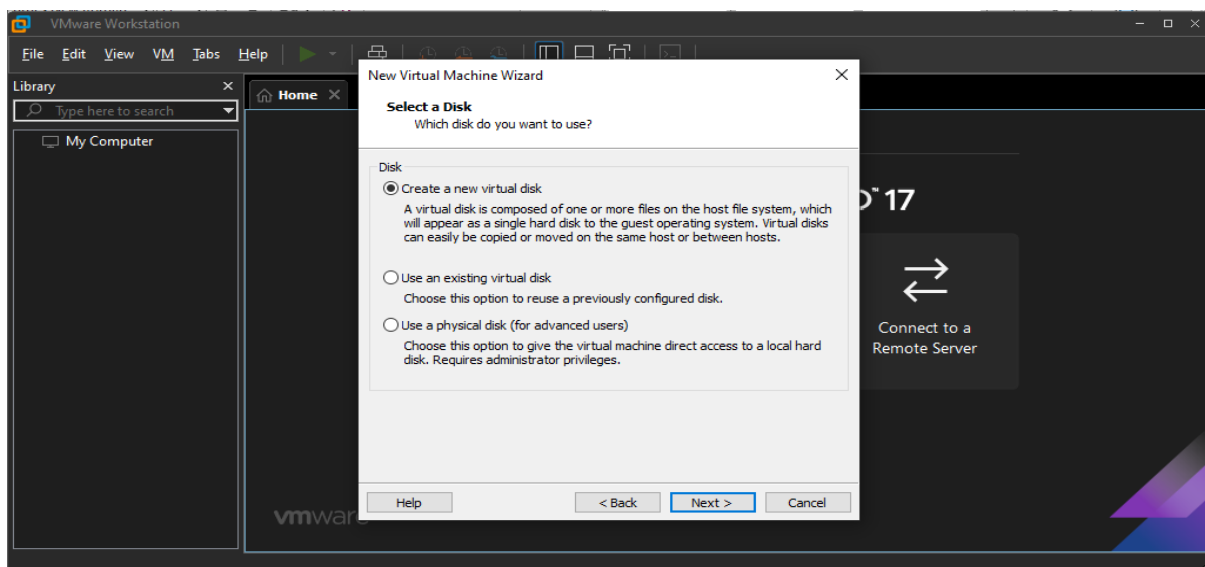
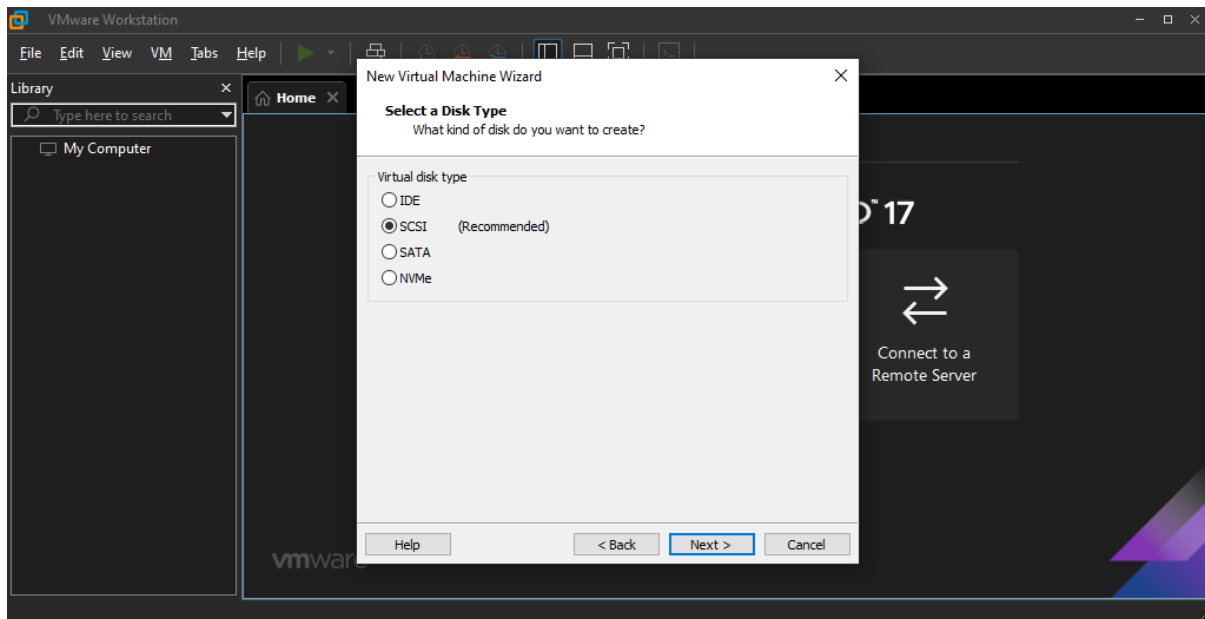
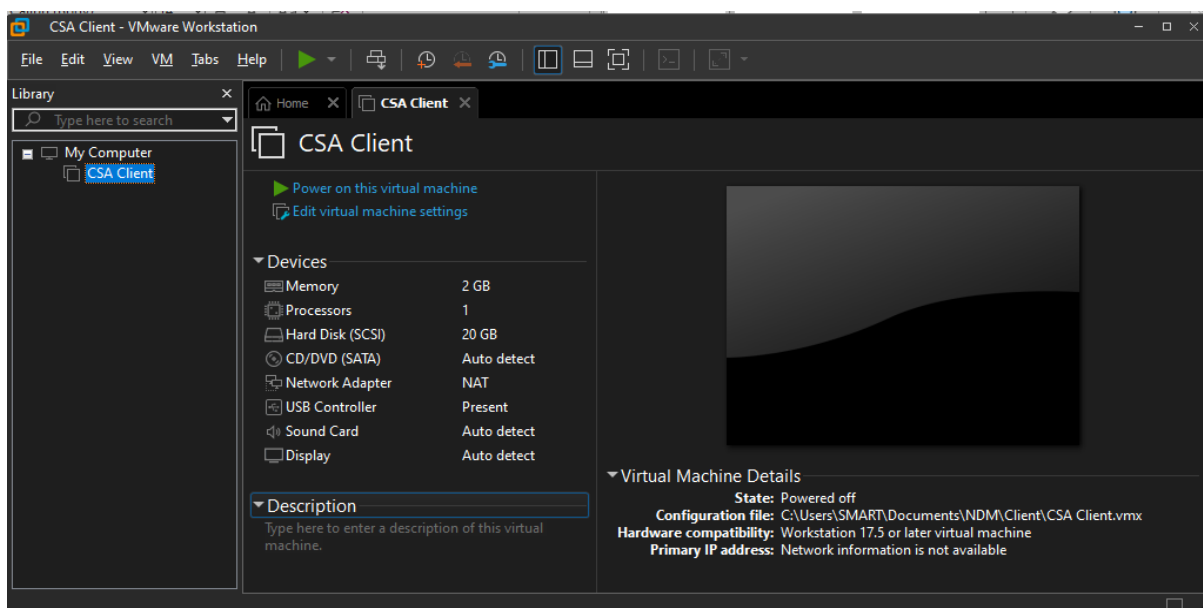
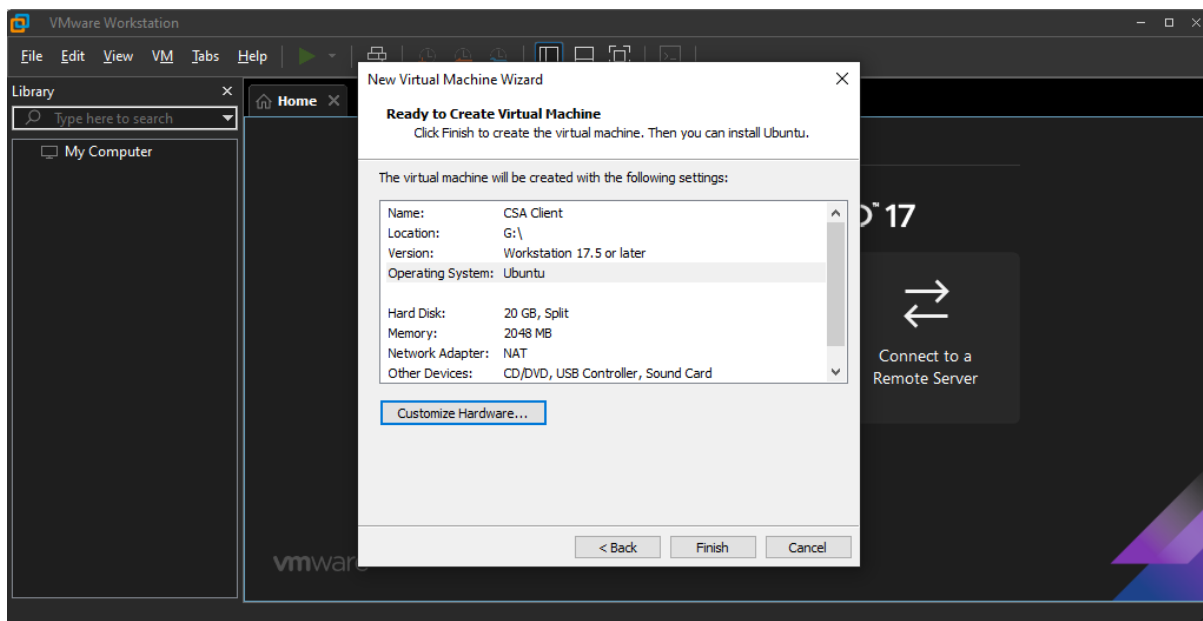
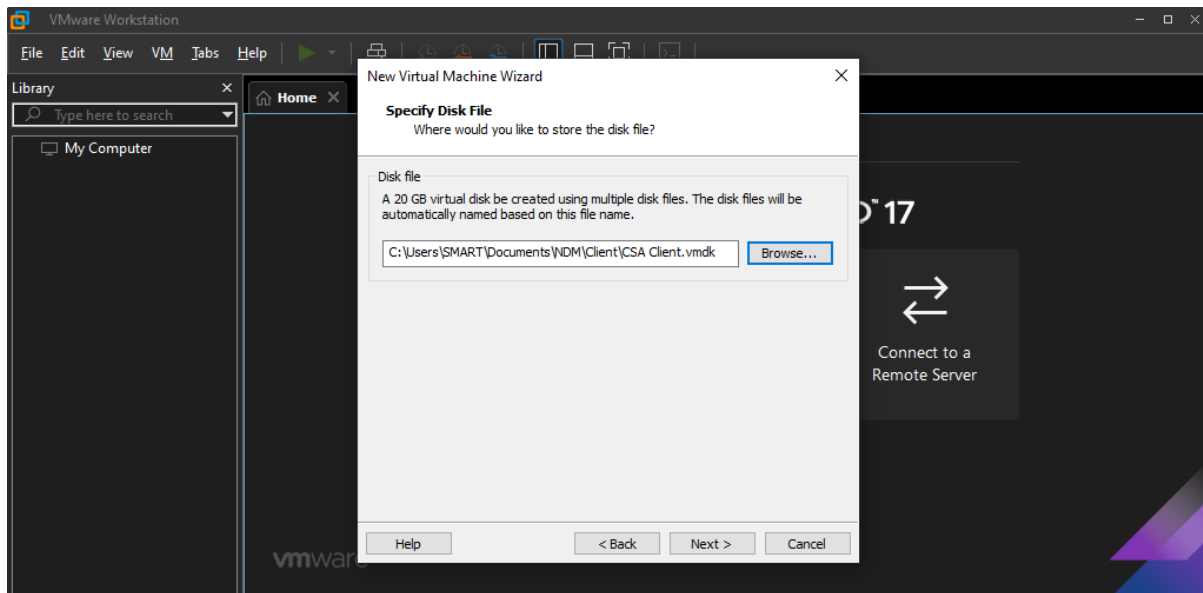


Figure 13









8. Under Network Adapter:
- Choose Custom
 - Select VMnet2.

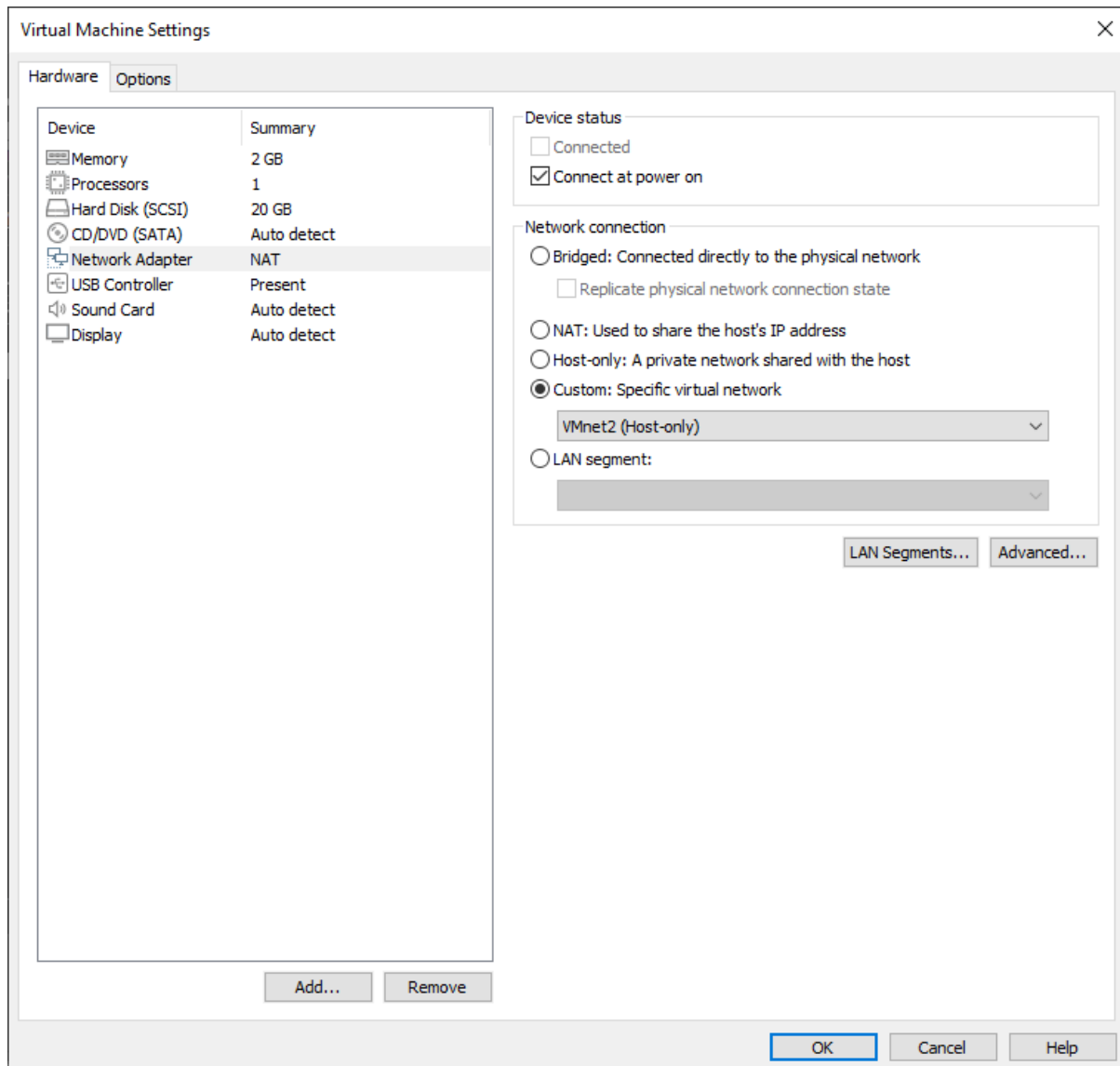


Figure 14 custom VMnet-2

9. Under CD/DVD settings:

- Select Use ISO Image File.
 - Attach Fedora ISO for client VM.
 - Attach CentOS ISO for server VM.
 - Save settings

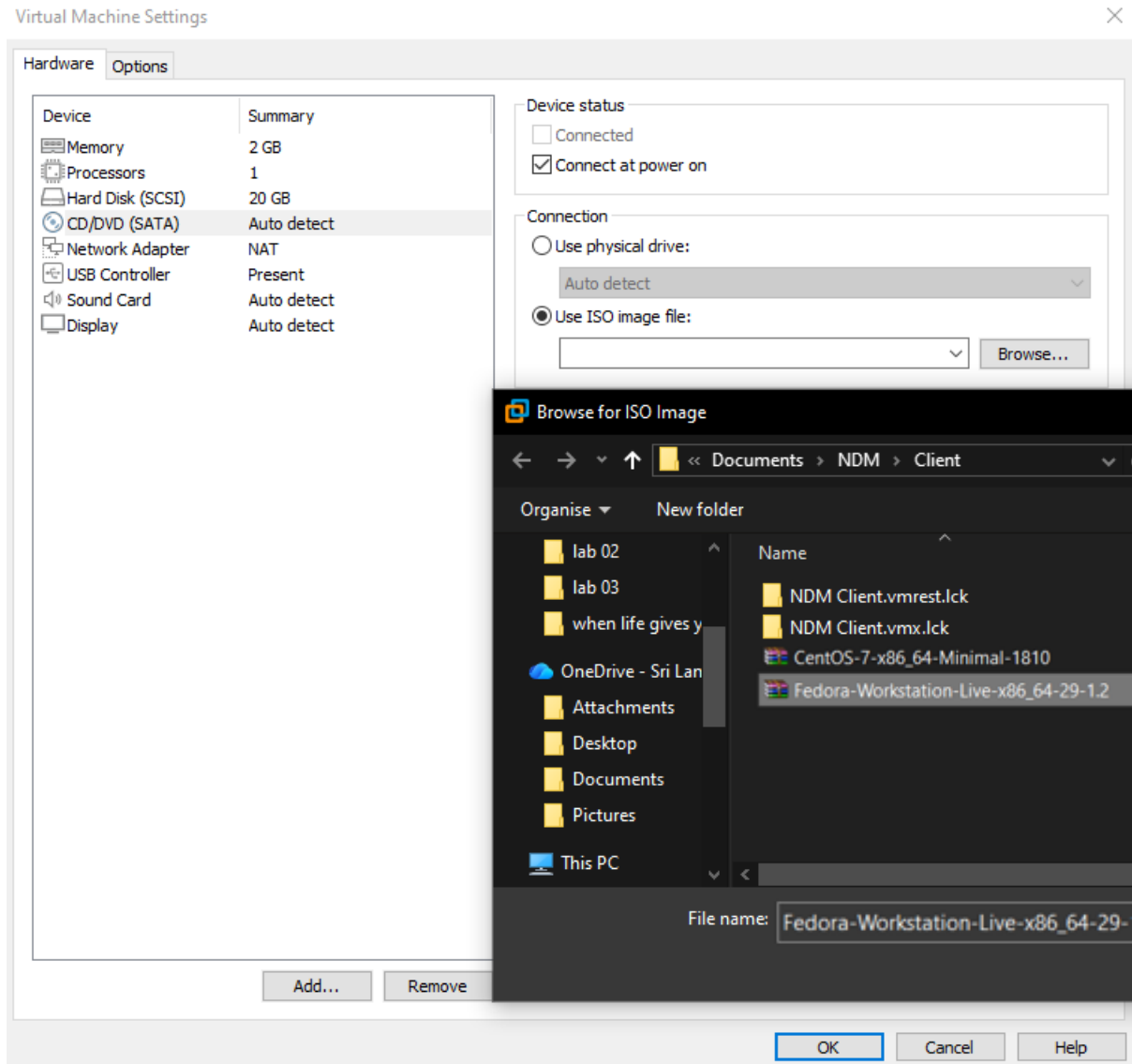
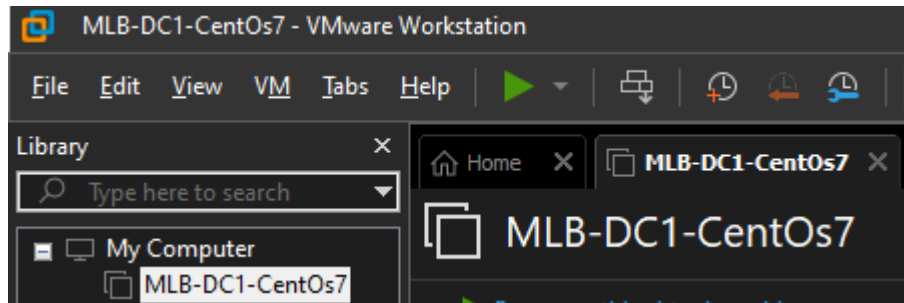


Figure 15 save

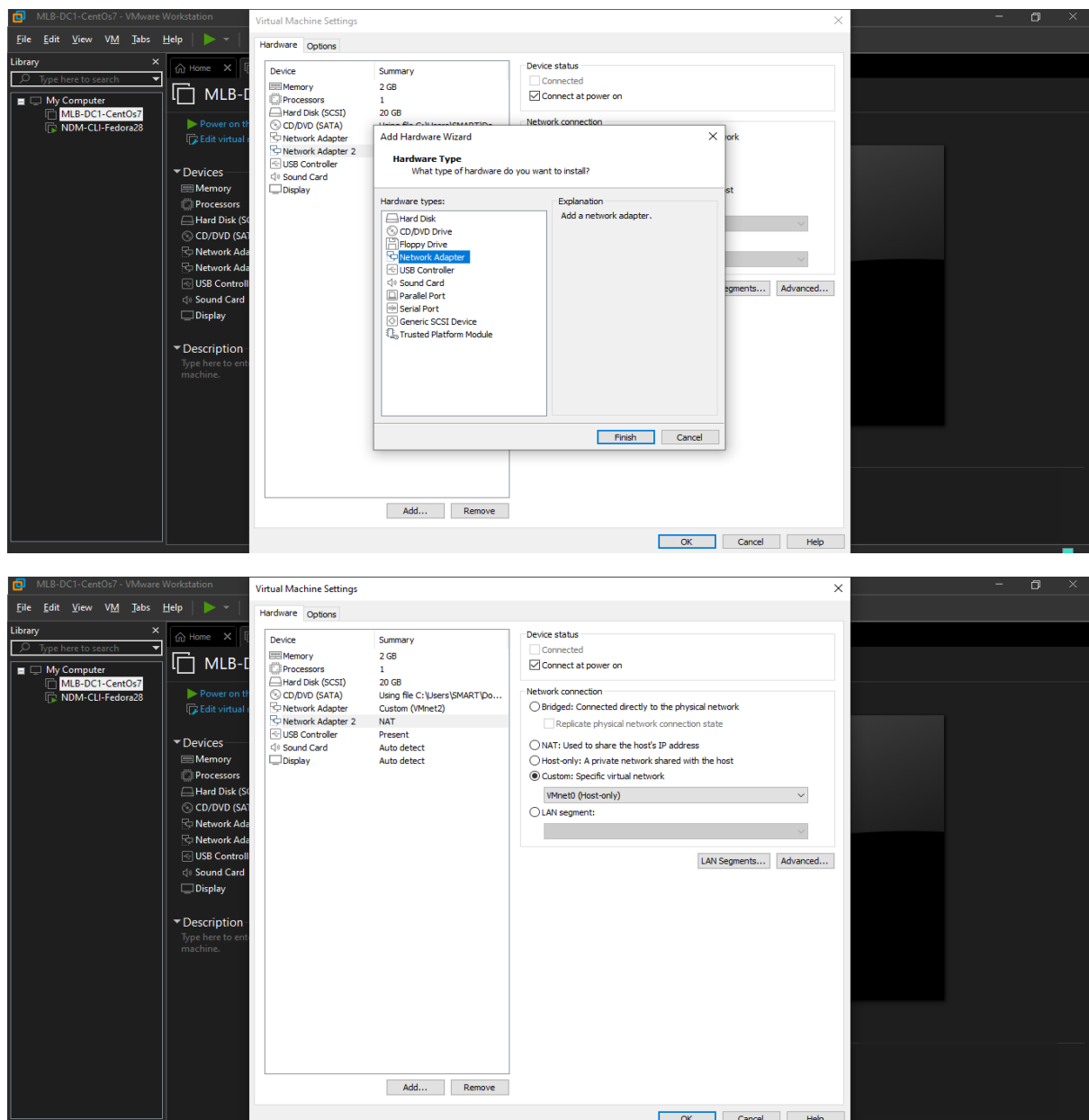
Lab sheet 01

Creating the Server – Specification

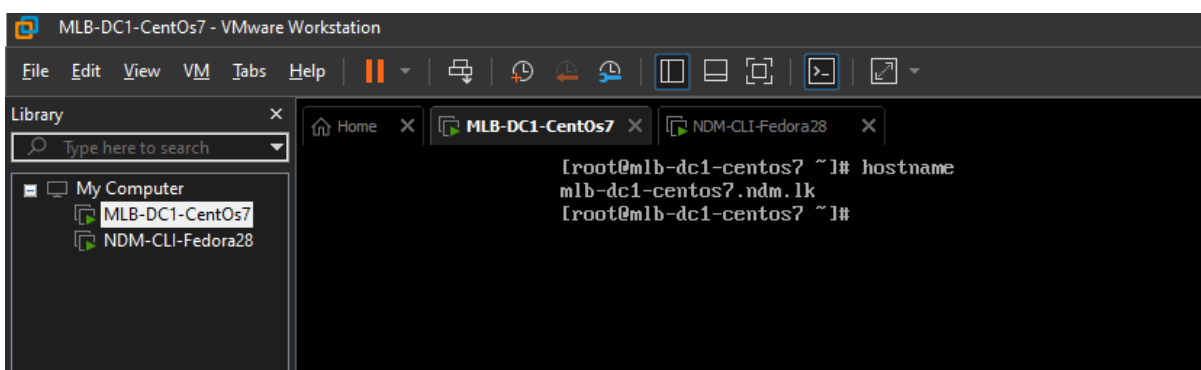
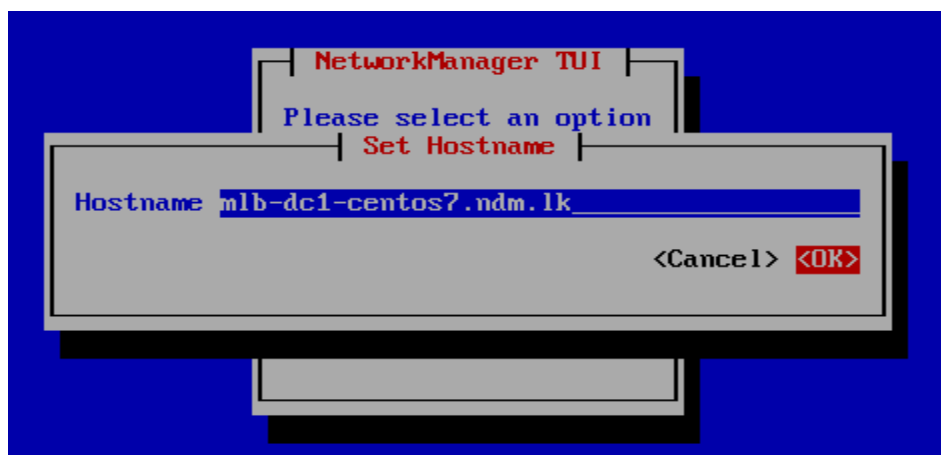
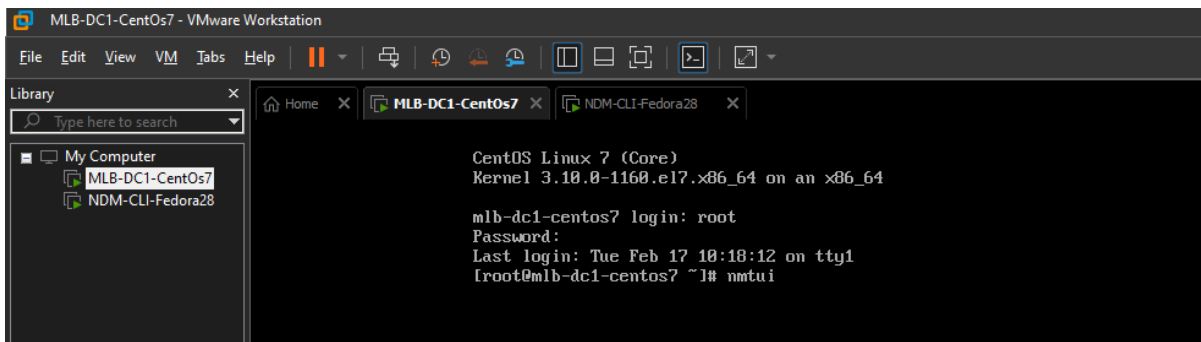
- VM Name/Machine Name: MLB-DC1-CentOs7



- Network Adapter 1: VMnet2
- Network Adapter 2: NAT

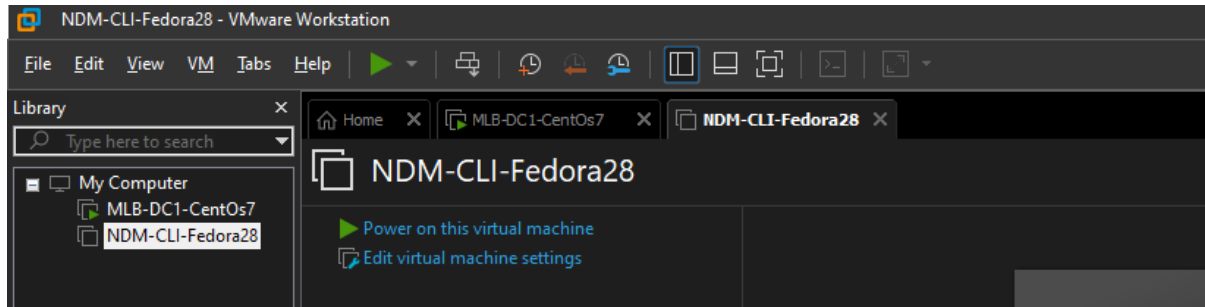


- Hostname (Under Network Settings): mlb-dc1-centos7.ndm.lk

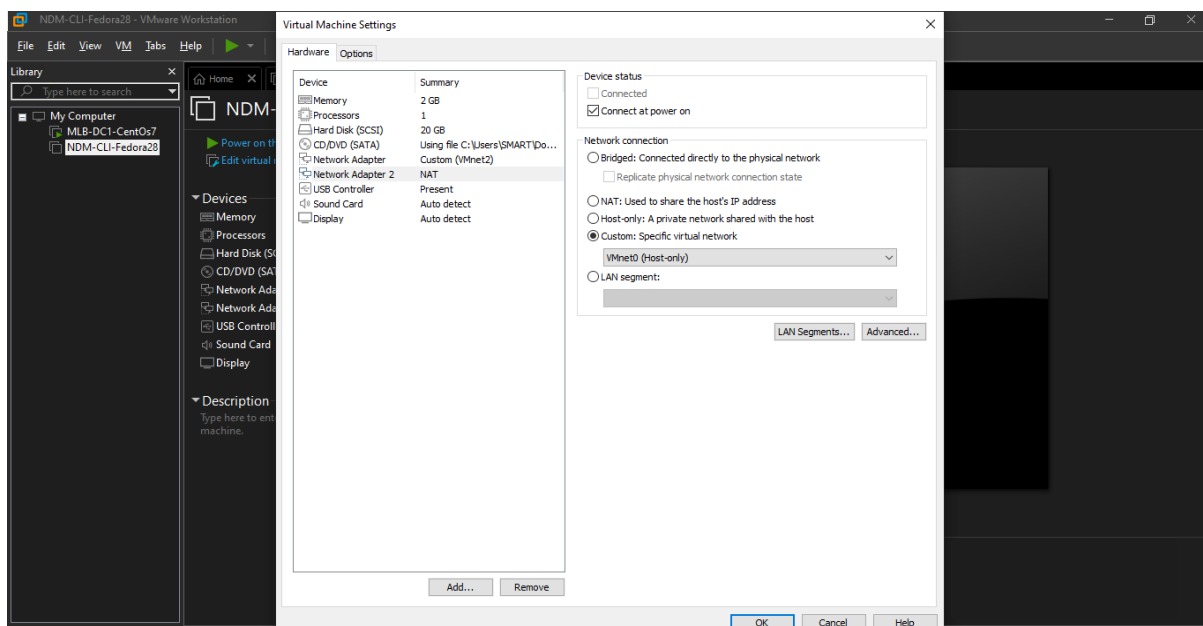
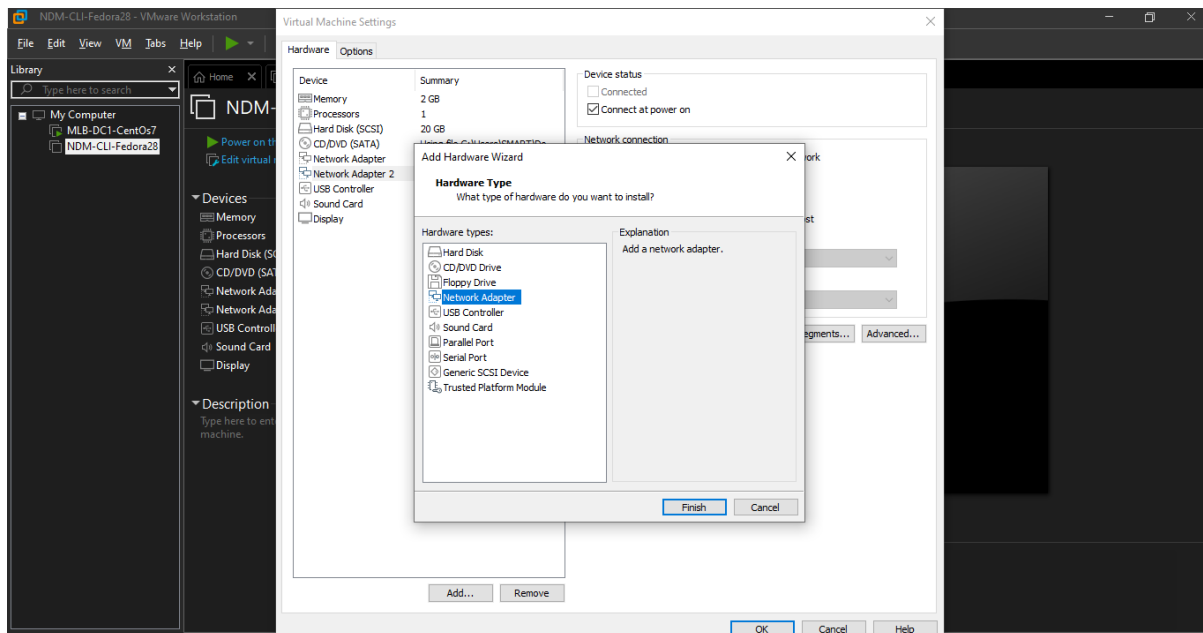


Creating the Client – Specifications

- VM Name/Machine Name: NDM-CLI-Fedora28



- Network Adapter 1: VMnet2
- Network Adapter 2: NAT



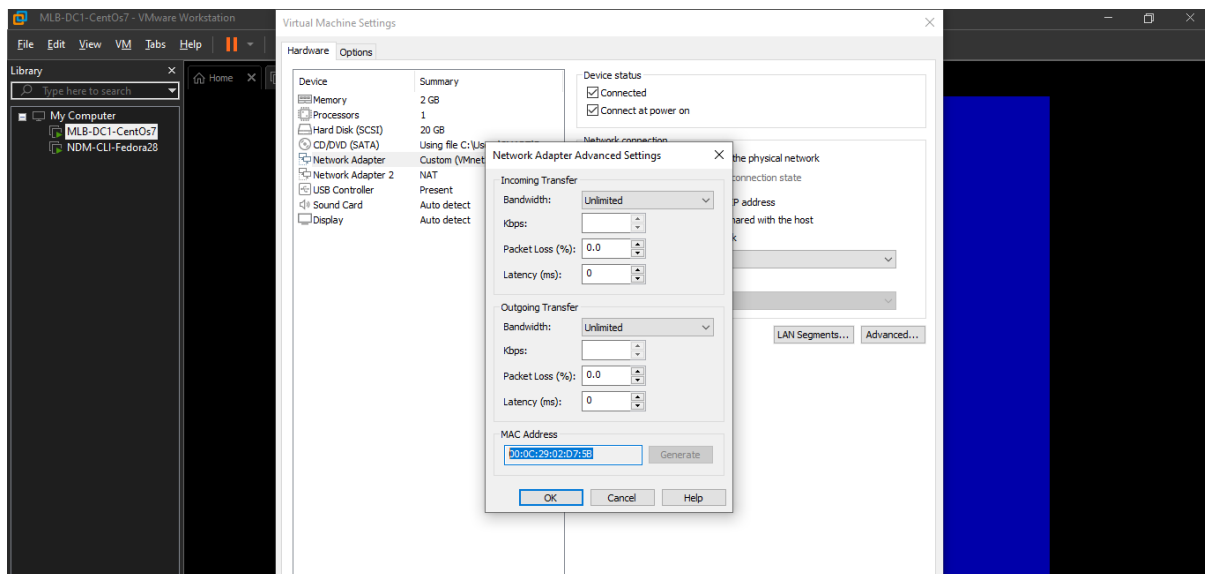
- Hostname (Under Network Settings): ndm-cli-fedora28.ndm.lk

```
client1@ndm-cli-fedora28:~  
[client1@ndm-cli-fedora28 ~]$ sudo hostname set-hostname ndm-cli-fedora28.ndm.lk  
[sudo] password for client1:
```

```
client1@ndm-cli-fedora28:~  
[client1@ndm-cli-fedora28 ~]$ hostname  
ndm-cli-fedora28.ndm.lk  
[client1@ndm-cli-fedora28 ~]$
```

Network Configurations

Server:



Edit Connection

Profile name Umet2
 Device 00:0C:29:02:D7:65 (ens37)

= ETHERNET <Show>

■ IPv4 CONFIGURATION <Manual>

Addresses 10.0.1.2/24 <Remove>
<Add...>

Gateway 10.0.1.1

DNS servers <Add...>

Search domains <Add...>

Routing (No custom routes) <Edit...>
☐ Never use this network for default route
☐ Ignore automatically obtained routes
☐ Ignore automatically obtained DNS parameters
☐ Require IPv4 addressing for this connection

■ IPv6 CONFIGURATION <Automatic>

☒ Automatically connect
☒ Available to all users

<Show>

<Cancel> <OK>

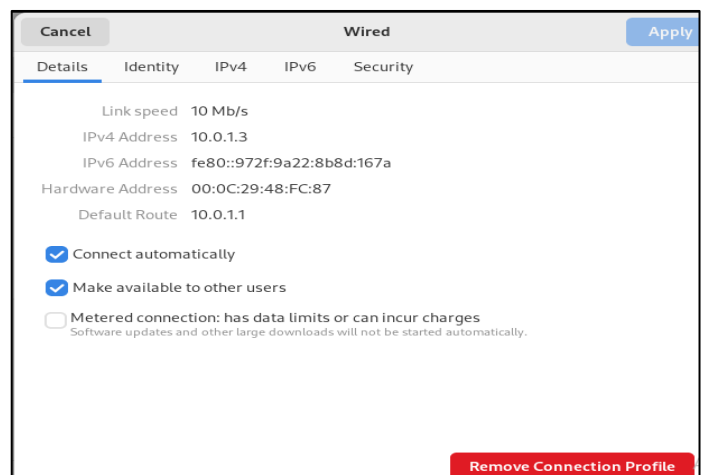
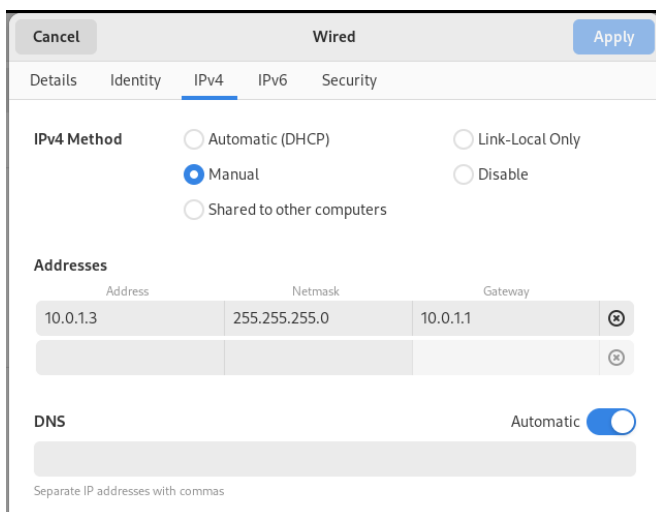
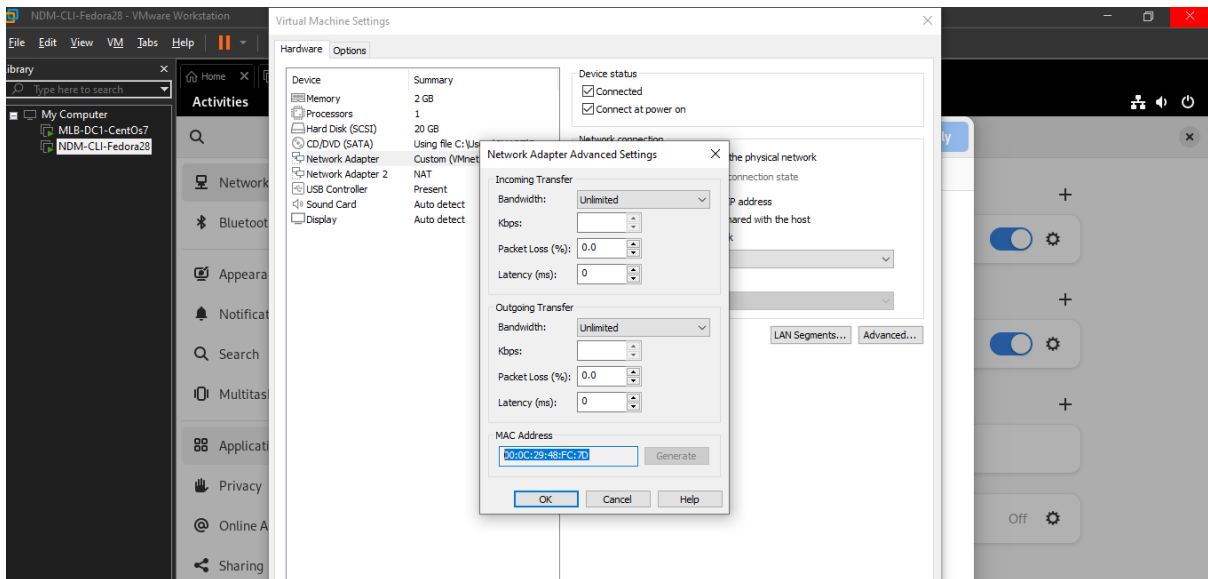
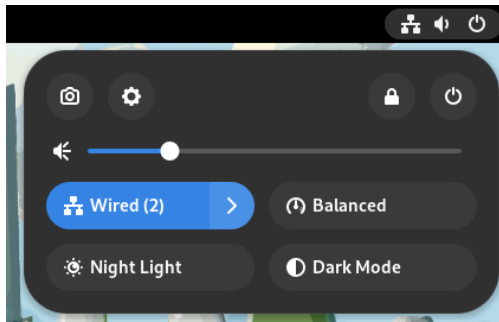
Ethernet (ens33) ■ <Deactivate>
 NAT

Ethernet (ens37)
* Umet2

<Back>

22

Client:



```
[root@mlb-dc1-centos7 ~]# ping 10.0.1.2
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=0.040 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=0.257 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=0.241 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=0.240 ms
64 bytes from 10.0.1.2: icmp_seq=5 ttl=64 time=0.246 ms
64 bytes from 10.0.1.2: icmp_seq=6 ttl=64 time=0.259 ms
^C
--- 10.0.1.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 4999ms
rtt min/avg/max/mdev = 0.040/0.213/0.259/0.080 ms
[root@mlb-dc1-centos7 ~]#
```

```
[root@mlb-dc1-centos7 ~]# ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.
64 bytes from 10.0.1.3: icmp_seq=1 ttl=64 time=0.818 ms
64 bytes from 10.0.1.3: icmp_seq=2 ttl=64 time=2.12 ms
64 bytes from 10.0.1.3: icmp_seq=3 ttl=64 time=2.01 ms
64 bytes from 10.0.1.3: icmp_seq=4 ttl=64 time=1.78 ms
^C
--- 10.0.1.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.818/1.686/2.126/0.515 ms
[root@mlb-dc1-centos7 ~]#
```


Conclusion

In this lab, it was effectively showcased how the efficient deployment of multiple operating systems on a single physical computer is enabled by virtualization platforms. Through the creation of Fedora and CentOS virtual machines connected via a virtual switch, fundamental concepts such as virtualization, operating system installation, and virtual networking were practically explored. These foundational skills are considered essential for careers in system administration, the development of testing environments, and cloud computing applications.

References

- Fedora 64-bit image:
https://download.fedoraproject.org/pub/fedora/linux/releases/37/Workstation/x86_64/iso/Fedora-Workstation-Live-x86_64-37-1.7.iso
- CentOS7 64-bit image:
http://centos.exellmedia.net/7.9.2009/isos/x86_64/CentOS-7-x86_64-Minimal-2207-02.iso
- How to create a VM in VMware: <https://www.youtube.com/watch?v=BHpRTVP8upg>
- How to create a VM in Oracle VirtualBox: https://www.youtube.com/watch?v=sB_5fqiySi4